



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Practical No. : 3

Aim –

Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required –

VSCode, Chrome, HTML5, CSS3, Javascript (ES6)

Experiment Program Code –

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Grading System</title>

<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Segoe UI,Arial,sans-serif;
}
body{
    background:#f6f8fc;
    color:#1f2937;
}
.page{
    min-height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    padding:40px;
}
.card{
    width:900px;
    max-width:100%;
    display:grid;
    grid-template-columns:320px 1fr;
    background:#fff;
    border-radius:22px;
    overflow:hidden;
    box-shadow:0 20px 50px rgba(0,0,0,.08);
}
```

```
}
.sidebar{
  background:#111827;
  color:#fff;
  padding:48px;
  display:flex;
  flex-direction:column;
  justify-content:space-between;
}
.sidebar h1{
  font-size:34px;
  line-height:1.15;
  margin:18px 0;
}
.sidebar p{
  color:#cbd5e1;
  line-height:1.7;
}
.main{
  padding:48px;
}
h2{
  font-size:28px;
  margin-bottom:8px;
}
.sub{
  color:#6b7280;
  margin-bottom:32px;
}
label{
  display:block;
  margin-bottom:8px;
  font-weight:600;
}
input{
  width:100%;
  padding:14px;
  border:1.5px solid #d8dee8;
  border-radius:12px;
  margin-bottom:22px;
  font-size:15px;
  transition:.25s;
}
input:focus{
  outline:none;
  border-color:#2563eb;
  box-shadow:0 0 4px rgba(37,99,235,.08);
}
button{
  width:100%;
  padding:15px;
  border:none;
  border-radius:12px;
  background:#111827;
  color:#fff;
  font-size:15px;
  font-weight:600;
  cursor:pointer;
  transition:.25s;
}
button:hover{
  background:#2563eb;
}
#reportPage{
  display:none;
}
```

```
.report{
  width:760px;
  max-width:100%;
  background:#fff;
  border-radius:22px;
  box-shadow:0 20px 50px rgba(0,0,0,.08);
  overflow:hidden;
}
.reportTop{
  height:8px;
  background:#2563eb;
}
.reportBody{
  padding:45px;
}
.grade{
  width:110px;
  height:110px;
  border-radius:50%;
  display:flex;
  align-items:center;
  justify-content:center;
  margin:0 auto;
  font-size:38px;
  font-weight:700;
  color:#fff;
}
.title{
  text-align:center;
  font-size:30px;
  margin-top:20px;
}
.msg{
  text-align:center;
  color:#6b7280;
  margin-top:8px;
  margin-bottom:35px;
}
table{
  width:100%;
  border-collapse:collapse;
}
td{
  padding:18px;
  border-bottom:1px solid #e2e8f0;
}
td:first-child{
  font-weight:600;
  color:#4b5563;
  width:35%;
}
.back{
  margin-top:35px;
  background:#111827;
}
.footer{
  width:100%;
  text-align:center;
  padding:18px;
  color:#6b7280;
  font-size:14px;
  letter-spacing:.3px;
}
.footer strong{
  color:#111827;
  font-weight:600;
}
```

```

}
@media(max-width:780px){
    .card{
        grid-template-columns:1fr;
    }
    .sidebar{
        display:none;
    }
}
</style>
</head>

<body>
<div id="formPage" class="page">
<div>
<div class="card">
<div class="sidebar">
<div>
<div style="font-size:42px">❌</div>
<h1>Student<br>Grading System</h1>
<p>Academic Performance Evaluation Portal</p>
</div>
<div style="color:#94a3b8;font-size:13px">
Department of Computer Science
</div>
</div>

<div class="main">
<h2>Evaluate Performance</h2>
<p class="sub">
Enter student information to generate the report.
</p>

<label>Student Name</label>
<input
type="text"
id="name"
placeholder="Enter student name">

<label>Marks (0-100)</label>
<input
type="number"
id="marks"
placeholder="Enter marks">

<button onclick="gradeSystem()">
Generate Report
</button>
</div>
</div>

<div class="footer">
Developed by <strong>Sanskruti Gorle</strong> | 24070521025
</div>
</div>
</div>

<div id="reportPage" class="page">
<div>
<div class="report">
<div id="bar" class="reportTop"></div>

<div class="reportBody">
<div id="gradeCircle" class="grade"></div>

<h2 class="title">

```

```
Student Grade Report
</h2>

<p id="performance" class="msg"></p>

<table>
<tr>
<td>Student Name</td>
<td id="rName"></td>
</tr>

<tr>
<td>Marks</td>
<td id="rMarks"></td>
</tr>

<tr>
<td>Grade</td>
<td id="rGrade"></td>
</tr>

<tr>
<td>Result</td>
<td id="rResult"></td>
</tr>
</table>

<button class="back" onclick="goBack()">
← Back
</button>
</div>
</div>

<div class="footer">
Developed by <strong>Sanskriti Gorle</strong> | 24070521025
</div>
</div>
</div>

<script>
function gradeSystem(){
var name=document.getElementById("name").value.trim();
var marks=document.getElementById("marks").value;

if(name==""){
alert("Please enter student name.");
return;
}

if(marks==""){
alert("Please enter marks.");
return;
}

marks=Number(marks);

if(marks<0||marks>100){
alert("Marks should be between 0 and 100.");
return;
}

var grade,result,msg,color;

if(marks>=90){
grade="A+";
result="PASS";
```

```

msg="Outstanding Performance";
color="#16a34a";
}
else if(marks>=80){
grade="A";
result="PASS";
msg="Excellent Performance";
color="#16a34a";
}
else if(marks>=70){
grade="B";
result="PASS";
msg="Very Good";
color="#2563eb";
}
else if(marks>=60){
grade="C";
result="PASS";
msg="Good Performance";
color="#f59e0b";
}
else if(marks>=50){
grade="D";
result="PASS";
msg="Satisfactory";
color="#f59e0b";
}
else if(marks>=40){
grade="E";
result="PASS";
msg="Needs Improvement";
color="#ea580c";
}
else{
grade="F";
result="FAIL";
msg="Better Luck Next Time";
color="#dc2626";
}

document.getElementById("formPage").style.display="none";
document.getElementById("reportPage").style.display="flex";

document.getElementById("gradeCircle").style.background=color;
document.getElementById("bar").style.background=color;

document.getElementById("gradeCircle").innerHTML=grade;
document.getElementById("performance").innerHTML=msg;
document.getElementById("rName").innerHTML=name;
document.getElementById("rMarks").innerHTML=makes+"/100";
document.getElementById("rGrade").innerHTML=grade;
document.getElementById("rResult").innerHTML="<b style='color:"+color+"'>"+result+"</b>";
}

function goBack(){
document.getElementById("reportPage").style.display="none";
document.getElementById("formPage").style.display="flex";
}
</script>

</body>
</html>

```

Output –

The screenshot shows a web browser window with the title "Student Grading System". The address bar shows the file path: C:/Users/SANSKRUTI/Downloads/Javascript/practical%203/grading%20system/index.html. The main content area features a dark blue sidebar on the left with the text "Student Grading System", "Academic Performance Evaluation Portal", and "Department of Computer Science". The main white area is titled "Evaluate Performance" with the instruction "Enter student information to generate the report." Below this, there are two input fields: "Student Name" with the value "Sanskriti" and "Marks (0-100)" with the value "96". A dark blue button labeled "Generate Report" is positioned below the marks field. At the bottom of the page, it says "Developed by Sanskruti Gorle | 24070521025". The Windows taskbar at the bottom shows the date as 01-08-2026 and time as 23:23.

Field	Value
Student Name	Sanskriti
Marks (0-100)	96

The screenshot shows the "Student Grade Report" page. It features a large green circle with "A+" at the top. Below it, the title "Student Grade Report" is followed by the subtitle "Outstanding Performance". A table displays the student's information:

Field	Value
Student Name	Sanskriti
Marks	96/100
Grade	A+
Result	PASS

At the bottom of the report, there is a dark blue button labeled "Back". The footer of the page states "Developed by Sanskruti Gorle | 24070521025". The Windows taskbar at the bottom shows the date as 01-08-2026 and time as 23:23.

Case Study Title –

To develop a login page using HTML and JavaScript that validates user credentials using conditional statements (if, else if, and else) and displays appropriate messages based on the input.

Case Study Program Code –

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Sign In</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="stylesheet">
<link rel="stylesheet" href="style.css">
</head>

<body>

<div class="container">

  <div class="login-card">

    <div class="logo">🔗</div>

    <h1>Welcome back</h1>
    <p>Sign in to continue</p>

    <form id="loginForm">

      <label>Email</label>
      <input
        type="email"
        id="email"
        placeholder="name@example.com"
        required
      >

      <div class="password-header">
        <label>Password</label>
        <a href="#">Forgot password?</a>
      </div>

      <div class="password-box">

        <input
          type="password"
          id="password"
          placeholder="Enter password"
          required
        >

        <button
          type="button"
          id="togglePassword">
          Show
        </button>

      </div>

      <small id="error"></small>

      <button
        class="login-btn"
        type="submit">
        Continue
      </button>

    </form>

  </div>

</div>
```

```
    </form>

    <div class="bottom">
      Don't have an account?
      <a href="#">Sign up</a>
    </div>

  </div>

</div>

<script src="script.js"></script>

</body>
</html>
```

style.css

```
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:'Inter',sans-serif;
}

body{
background:#f6f7fb;
display:flex;
justify-content:center;
align-items:center;
height:100vh;
}

.container{
width:100%;
display:flex;
justify-content:center;
padding:20px;
}

.login-card{
width:380px;
background:#fff;
padding:40px;
border-radius:18px;
box-shadow:0 10px 35px rgba(0,0,0,.08);
border:1px solid #e5e7eb;
}

.logo{
width:50px;
height:50px;
background:#111827;
color:#fff;
border-radius:14px;
display:flex;
justify-content:center;
align-items:center;
font-size:22px;
margin:auto;
margin-bottom:25px;
}
```

```
}

h1 {
text-align:center;
font-size:28px;
margin-bottom:8px;
}

p{
text-align:center;
color:#6b7280;
margin-bottom:28px;
}

label{
display:block;
font-size:14px;
font-weight:600;
margin-bottom:8px;
}

input{
width:100%;
padding:13px 15px;
border:1px solid #d1d5db;
border-radius:10px;
font-size:15px;
outline:none;
transition:.3s;
}

input:focus{
border-color:#111827;
box-shadow:0 0 3px rgba(17,24,39,.08);
}

.password-header{
display:flex;
justify-content:space-between;
align-items:center;
margin-top:20px;
margin-bottom:8px;
}

.password-header a{
font-size:13px;
text-decoration:none;
color:#111827;
}

.password-box {
position:relative;
}

#togglePassword{
position:absolute;
right:12px;
top:50%;
transform:translateY(-50%);
border:none;
background:none;
cursor:pointer;
font-weight:600;
color:#555;
}

.login-btn{
```

```
width:100%;
margin-top:25px;
padding:14px;
border:none;
background:#111827;
color:#fff;
font-size:15px;
font-weight:600;
border-radius:10px;
cursor:pointer;
transition:.3s;
}

.login-btn:hover{
background:#000;
}

.bottom{
margin-top:25px;
text-align:center;
font-size:14px;
color:#6b7280;
}

.bottom a{
color:#111827;
text-decoration:none;
font-weight:600;
}

#error{
display:block;
margin-top:8px;
color:#dc2626;
font-size:13px;
min-height:18px;
}
```

script.js

```
const password = document.getElementById("password");
const toggle = document.getElementById("togglePassword");
const form = document.getElementById("loginForm");
const error = document.getElementById("error");

toggle.onclick = function () {

  if (password.type === "password") {
    password.type = "text";
    toggle.innerHTML = "Hide";
  }
  else {
    password.type = "password";
    toggle.innerHTML = "Show";
  }
}

form.onsubmit = function (e) {

  let pass = password.value;
```

```
let hasUpper = false;
let hasLower = false;
let hasNumber = false;
let hasSpecial = false;

for (let i = 0; i < pass.length; i++) {

    let ch = pass[i];

    if (ch >= 'A' && ch <= 'Z') {
        hasUpper = true;
    }
    else if (ch >= 'a' && ch <= 'z') {
        hasLower = true;
    }
    else if (ch >= '0' && ch <= '9') {
        hasNumber = true;
    }
    else {
        hasSpecial = true;
    }
}

if (pass.length < 8) {
    e.preventDefault();
    error.innerHTML = "Password must be at least 8 characters.";
}

else if (!hasUpper) {
    e.preventDefault();
    error.innerHTML = "Password must contain an uppercase letter.";
}

else if (!hasLower) {
    e.preventDefault();
    error.innerHTML = "Password must contain a lowercase letter.";
}

else if (!hasNumber) {
    e.preventDefault();
    error.innerHTML = "Password must contain a number.";
}

else if (!hasSpecial) {
    e.preventDefault();
    error.innerHTML = "Password must contain a special character.";
}

else {
    error.innerHTML = "";
    alert("Login Successful!");
}
}
```

Output —

Sign In

C:/Users/SANSKRUTI/Downloads/Javascript/practical%203/login%20validation/index.html



Welcome back

Sign in to continue

Email

Password [Forgot password?](#)

 [Show](#)

Password must contain a special character.

Continue

Don't have an account? [Sign up](#)

27°C Partly cloudy

Search

ENG IN

23:26 01-08-2026

Sign In

C:/Users/SANSKRUTI/Downloads/Javascript/practical%203/login%20validation/index.html



Welcome back

Sign in to continue

Email

Password [Forgot password?](#)

 [Show](#)

Password must contain a number.

Continue

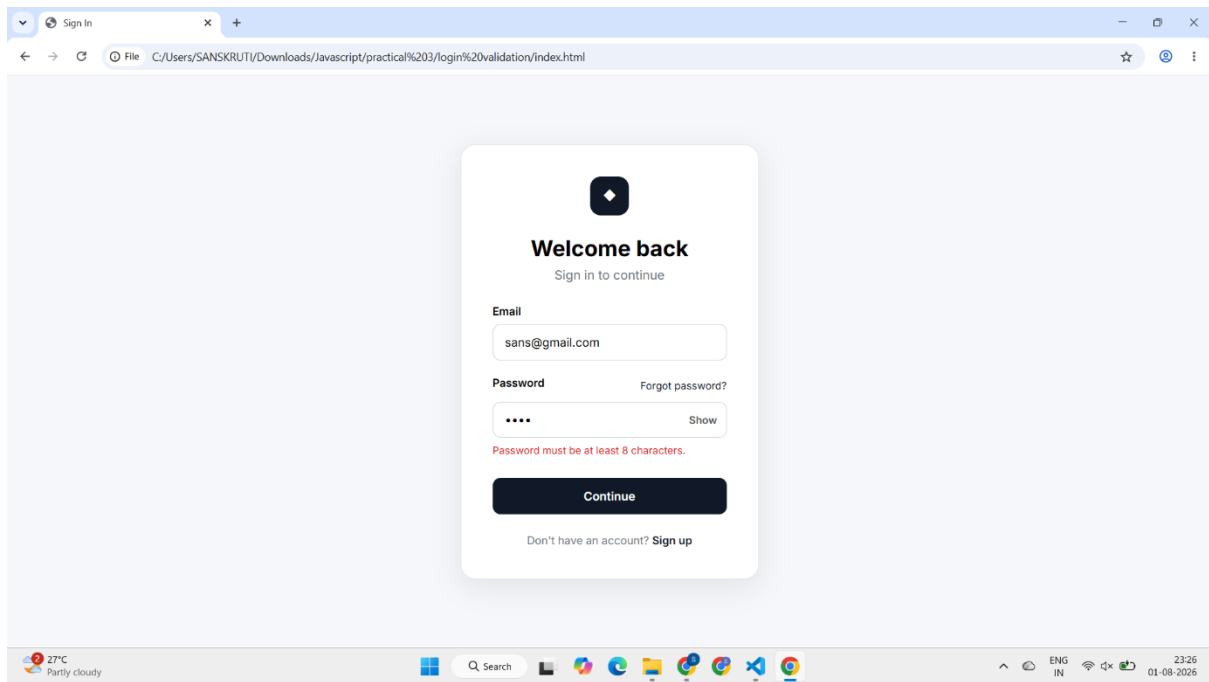
Don't have an account? [Sign up](#)

27°C Partly cloudy

Search

ENG IN

23:26 01-08-2026



Conclusion –

The experiment was successfully performed to understand JavaScript conditional statements and form validation. A Student Grading System was developed to assign grades on the marks entered by the user. A Login Page was also created to validate user credentials using conditional statements and display appropriate messages for invalid input.